



Prerequisites

Team Oriented Project

Group A

Linux Kernel Extension for DM-Verity Authentication, Verification and Setup

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Cooperation with:

**BMW
GROUP**



ROLLS-ROYCE
MOTOR CARS LTD

[BMW Car IT, Ulm](#)

Eva-Helena Zorman

Tobias Kaufmann

Philipp Graf

This guide outlines the necessary software packages required for building, securing, and emulating the system environment. All packages are installed using the apt package manager.

1. Core Development and Build Tools

These packages provide the essential utilities needed for compiling source code and fetching resources.

- The process begins by executing `sudo apt update` to refresh the package index.
- The primary tools are installed with `sudo apt install -y build-essential git wget curl ca-certificates pkg-config libssl-dev`. This includes the GNU compiler collection (build-essential), version control (git), network utilities (wget, curl), and the SSL development libraries (libssl-dev), which are necessary for cryptographic functions.

2. QEMU Virtualization Environment

QEMU provides the necessary tools for creating the virtual machine (VM) and managing its disk images.

- The command `sudo apt install -y qemu-system-x86 qemu-utils` installs the required QEMU components.
- This provides the “qemu-system-x86_64” emulator for running the virtual machine, and the `qemu-img` utility for creating, converting, and managing virtual disk images.

3. Root Filesystem and EXT4 Management

These packages are required for constructing the minimal operating system image and manipulating the Ext4 filesystem format.

- The necessary tools are installed with `sudo apt install -y debootstrap e2fsprogs rsync`.
- This package set includes **debootstrap**, which is used for bootstrapping a minimal Debian/Ubuntu root filesystem structure.
- It also provides the **e2fsprogs** suite (e.g., `mkfs.ext4`, `resize2fs`, `e2fsck`), essential for Ext4 filesystem creation and repair, along with **rsync** for efficient file transfer into the disk image.

4. DM-Verity and Cryptsetup Utilities

These specialized tools are essential for implementing the device mapper verity security features.

- The utilities are installed via `sudo apt install -y cryptsetup veritysetup`.

- This includes **veritysetup**, the command-line interface tool specifically used to create, format, and inspect **dm-verity** hash trees.
- The package also provides **cryptsetup**, which serves as the core backend library and command-line interface for the device mapper.

5. Loopback and Block Device Management

These utilities enable the manipulation and inspection of disk images as virtual block devices.

- The relevant standard utilities are installed with `sudo apt install -y util-linux`.
- This includes **losetup**, which is used for setting up and managing loop devices, a requirement for treating disk image files as block devices.
- It also provides **blockdev**, a tool used to query or set parameters of block devices.

6. Cryptography and Signing Tools

The OpenSSL package provides the cryptographic primitives required for key management and digital signing operations.

- The installation command is `sudo apt install -y openssl`.
- **Usage:** OpenSSL is used for generating the PKCS#7 signature applied to the 196-byte metadata header, and for creating the required key pair (`bl_private.pem` and `bl_cert.pem`) necessary for the image signing process.

Installation Commands Summary

For a quick reference, execute the following commands in order:

```
# 1. Update and basic build tools
sudo apt update

sudo apt install -y build-essential git wget curl ca-certificates pkg-config libssl-dev

# 2. QEMU
sudo apt install -y qemu-system-x86 qemu-utils

# 3. Root filesystem and ext4 tools
sudo apt install -y debootstrap e2fsprogs rsync

# 4. dm-verity and cryptsetup tools
sudo apt install -y cryptsetup veritysetup

# 5. Loopback and block device utilities
sudo apt install -y util-linux

# 6. Crypto & signing tools
sudo apt install -y openssl
```